Convex Optimization and Approximation Algorithms
Problem Set 4 – CS 6515/4540 (Fall 2026)

This problem set is due on **Sunday, October 25th**. Submission is via Gradescope. Your solution must be a typed pdf (e.g. via LaTeX) – no handwritten solutions.

16 Non-Convex Function

In class we assumed function f is convex. This problem considers the non-convex case. We want to show that for β-smooth f, after T iterations with step size $\eta \leq 1/\beta$ we can find a point x' with

$$\|\nabla f(x')\| \leq \sqrt{\frac{2}{\eta \cdot t}(f(x^0) - f(x^*))}.$$

(Note that for a local optimum we have $\nabla f(x) = 0$, so a small norm $\|\nabla f(x')\|$ indicates that we are close to a local optimum or saddle point.)

Proving this from scratch is a bit tricky, so we provide the following subproblems to guide you to a proof. Each subproblem can be solved in a few lines of calculation/algebra.

Problem:

1. Show $f(x^{t+1}) \leq f(x^t) - \frac{\eta}{2}\|\nabla f(x^t)\|^2$ (Hint: Check the proof from class. Does it work for non-convex functions?)

   This is exactly the same as the proof from class. First, note that if a function is β-smooth, then it is also L-smooth for any $L \geq \beta$ (this is clear from the definition). So, since $0 < \eta \leq 1/\beta$, we know f is $(1/\eta)$-smooth.

   Now, we will use the quadratic upper bound of smooth functions. We use $x = x^t$ and $y = x^{t+1}$. Note that by definition, $x^{t+1} - x^t = -\eta \nabla f(x^t)$.

   Now,

   $$\begin{aligned}
   f\left(x^{t+1}\right) &\leq f\left(x^t\right) + \left\langle \nabla f\left(x^t\right), x^{t+1} - x^t \right\rangle + \frac{1}{2\eta}\left\|x^{t+1} - x^t\right\|^2 \\
   &= f\left(x^t\right) + \left\langle \nabla f\left(x^t\right), -\eta \nabla f\left(x^t\right) \right\rangle + \frac{1}{2\eta}\left\|-\eta \nabla f\left(x^t\right)\right\|^2 \\
   &= f\left(x^t\right) - \eta\left\langle \nabla f\left(x^t\right), \nabla f\left(x^t\right) \right\rangle + \frac{\eta^2}{2\eta}\left\|\nabla f\left(x^t\right)\right\|^2 \\
   &= f\left(x^t\right) - \eta\left\|\nabla f\left(x^t\right)\right\|^2 + \frac{\eta}{2}\left\|\nabla f\left(x^t\right)\right\|^2 \\
   &= f\left(x^t\right) - \frac{\eta}{2}\left\|\nabla f\left(x^t\right)\right\|^2
   \end{aligned}$$

   □

2. Show $\sum_{k=0}^{t} \|\nabla f(x^k)\|^2 \leq \frac{2}{\eta}(f(x^0) - f(x^*))$ (Hint: 1. implies $\frac{\eta}{2}\|\nabla f(x^t)\|^2 \leq ...$)

Rearranging part 1, we get

$$\left\|\nabla f\left(x^t\right)\right\|^2 \leq \frac{2}{\eta}\left(f\left(x^t\right) - f\left(x^{t+1}\right)\right)$$

for any t. Now, if we apply this inequality to every element of the summation on the left and cancel out the telescoping terms (formally we would use induction to show this is true), we get

$$\sum_{k=0}^{t}\left\|\nabla f\left(x^k\right)\right\|^2 \leq \frac{2}{\eta}\left(f\left(x^0\right) - f\left(x^{t+1}\right)\right)$$

Since $f\left(x^{t+1}\right) \geq f(x^*)$ by definition of x^*, this gives

$$\sum_{k=0}^{t}\left\|\nabla f\left(x^k\right)\right\|^2 \leq \frac{2}{\eta}\left(f\left(x^0\right) - f\left(x^*\right)\right)$$

□

3. Show $\min_{k=0\ldots t}\|\nabla f(x^k)\| \leq \sqrt{\frac{2}{\eta \cdot T}(f(x^0) - f(x^*))}$.

First,

$$\sum_{k=0}^{t}\left\|\nabla f\left(x^k\right)\right\|^2 \geq (t+1)\min_{k=0..t}\left\|\nabla f\left(x^k\right)\right\|^2$$

(this is true for any summation of values, we can only decrease each element in the summation).

If we go up to some t, that is $t+1$ iterations total, so we can define $T = t+1$.

So, we can use this and part 2, divide both sides by T, and take a square root. Getting the desired inequality. Note that since square root is monotonic on its domain, taking the square root of a min is the same as taking the min of square roots.

where x^* is the global optimum $f(x^*) = \min_x f(x)$.

You are allowed to use subproblems to solve later subproblems (e.g., use 1+2 to solve 3), even if you did not prove them.

17 Greedy Approximation Algorithm

Consider placing n items into m bins. Item i has weight w_i. We want to place these n items into the bins such that the maximum total weight of a bin is minimized. Consider the following greedy algorithm:

Sort the weights w_i in decreasing order. Going through the items in decreasing order of weight, we place each item in the bin with the least current load (as we iterate through the items and place more items into bins, the least loaded bin may change). Show this algorithm gives a 1.5−approximation for the least possible maximum load of any placement of these n items into the m bins.

Assume the elements are already in sorted order. Also, we will assume all the weights are positive, because we can ignore items with 0 weight (and negative weights could mean our algorithm isn't a 1.5-approximation).

We will consider 3 cases. The first case is that some heaviest bin (according to our algorithm) has at most one element. In this case, our algorithm actually gives the optimal solution, since that item must go in some bin.

The second case is that some heaviest bin contains exactly two elements. Let the first element we add be element i, and the second be element j. So, our bin has weight $w_i + w_j$. We can see that our algorithm will always start by adding one element to each bin, since weight 0 is always the smallest weight, so $j > m$, meaning there are at least $m+1$ elements with weight at least w_j. So, by the pigeonhole principle, any placement of items into bins will have at least one bin with weight at least $2w_j$. So, we know any solution gives maximum weight at least $\max\{2w_j, w_i\}$ (we just showed the $2w_j$ thing, and the w_i thing is trivial because item i has to go in some bin).

We can consider the two subcases. If $w_i \geq 2w_j$, then our solution gives weight $w_i + w_j \leq 1.5w_i$ and the best solution has max weight at least w_i, so our solution is a 1.5-approximation. If $2w_j \geq w_i$, then our solution gives weight $w_i + w_j \leq 3w_j$, and any solution has max weight at least $2w_j$, so this is still a 1.5-approximation.

Now, we consider the third and final case, that some heaviest bin contains at least 3 elements. Let W be the weight of this bin immediately before the final element was added, and let the final element added be element j. Since the elements were added in sorted order and there are at least three elements in this bin, we know $W \geq$ `sum of the first two elements added to this bin` $\geq 2w_j$, so the final max weight $W + w_j \leq 1.5W$. Also, because of how our algorithm works, when we added item j to this bin, every bin had weight at least W. So, the total sum of all of the weights of all elements is at least mW, so the true optimum of this problem must be at least W. Since our final max weight is at most $1.5W$, this means we are a 1.5-approximation in this case as well.

18 Hitting Set via Deterministic Rounding

Let $W = \{w_1, w_2, \ldots, w_n\}$ be a set of positive numbers and let $B_1, B_2, \ldots, B_m$ be subsets of the set $[n] = \{1, 2, \ldots, n\}$. We said $H \subseteq [n]$ is a *hitting set* if $H \cap B_i \neq \emptyset$ for all $1 \leq i \leq m$. Denote by $b = \max |B_j|$ the size of the largest set B_j, $1 \leq j \leq m$. Consider the `Minimum Hitting Set` problem, where we assign weight w_i to number i and aim to find the hitting set H of minimum weight $\sum_{i \in H} w_i$.

1. Write a LP instance for this problem. Clearly define the variables, objective function, and constraints.

The variables are x_i for $i \in \{1, ..., n\}$

Now, we want to minimize the cost function

$$\sum_{i=1}^{n} w_i x_i$$

under the constraints

$$\sum_{i \in B_j} x_i \geq 1 \quad \forall j$$

$$x_i \geq 0 \quad \forall i$$

This is the relaxed LP version, for the ILP, we would have $x_i \in \{0, 1\}$ $\forall i$.

2. Let x^* be a solution of the relaxed LP problem. Consider the deterministic rounding

$$\widehat{x}_i = \begin{cases} 1 \text{ if } x_i^* \geq b^{-1} \\ 0 \text{ else} \end{cases}$$

Prove that the set $H' = \{i \mid \widehat{x}_i = 1\}$ is a hitting set.

Consider any set B_j. We know that

$$\sum_{i \in B_j} x_i^* \geq 1$$

by the constraints of the LP. This means

$$|B_j| \max_{i \in B_j} x_i^* \geq 1$$

and since $|B_j| \leq b$, this means

$$\max_{i \in B_j} x_i^* \geq b^{-1}$$

So, for an i that maximizes this left side, we know $\widehat{x}_i = 1$, so B_j is hit by H'. This means every B_j is hit, so this is a hitting set.

3. Show that the deterministic rounding in part (b) yields a $b-$approximation of the `Minimum Hitting Set` problem.

Let LP^* be the optimal value to this LP. Note

$$LP^* = \sum_{i=1}^{n} w_i x_i^*$$

Now, note

$$\widehat{x}_i \leq b x_i^*$$

We can see this is true by checking both cases. If $\widehat{x}_i = 1$, the we know it's true, and it's clearly true if $\widehat{x}_i = 0$.
So,

$$\sum_{i=1}^{n} w_i \widehat{x}_i \leq b \sum_{i=1}^{n} w_i x_i^* = b \cdot LP^*$$

Also, note that since the LP is a relaxed version of the minimum hitting set problem, its optimum is at least as good, meaning LP^* is less than or equal to the optimum for the minimum hitting set problem.
So,

$$\sum_{i=1}^{n} w_i \widehat{x}_i \leq b \cdot \texttt{MininumHittingSetSolution}$$